Model 1: $\text{logit}\{P(Y_i=1)\} = \beta + b_{h_i} - b_{a_i}$

M teams with (transformed) ELOs b: $b_j \sim N(0, \tau)$ iid

N games with outcome Y: $Y_i \mid b_{h_i}, b_{a_i} \sim \text{Bernoulli}(p_i)$

$$p_i = \text{expit}(\beta + b_{h_i} - b_{a_i})$$

$$\mathcal{L}(\beta, \tau \mid Y, b) = f(Y, b) = f(Y \mid b) f(b)$$

$$f(Y \mid b) = \prod_{i=1}^{N} p_i^{Y_i} (1-p_i)^{1-Y_i}$$

$$f(b) = \{2\pi\tau\}^{-M/2} \exp\left\{-\frac{b^T b}{2\tau}\right\}$$

$$l = \log \mathcal{L} = \sum_{i=1}^{N} \{Y_i \log p_i + (1-Y_i)\log(1-p_i)\} - \frac{M}{2}\log(2\pi\tau) - \frac{b^T b}{2\tau}$$

$$\frac{\partial l}{\partial \beta} = \sum_{i=1}^{N}\left[\frac{Y_i}{p_i}\frac{\partial p_i}{\partial \beta} - \frac{1-Y_i}{1-p_i}\frac{\partial p_i}{\partial \beta}\right]$$

$$\text{expit}'(x) = \text{expit}(x)[1-\text{expit}(x)]$$

$$= \sum_{i=1}^{N} Y_i(1-p_i) - (1-Y_i)p_i = \sum_{i=1}^{N} Y_i - p_i$$

$$\frac{\partial l}{\partial \tau} = -\frac{M}{2\tau} + \frac{b^T b}{2\tau^2} \Rightarrow \hat{\tau}_{MLE} = \frac{b^T b}{M}$$

$$\frac{\partial l}{\partial b_j} = \left[\sum_{i:\, j \in \{h_i, a_i\}} \frac{Y_i}{p_i}\frac{\partial p_i}{\partial b_j} - \frac{1-Y_i}{1-p_i}\frac{\partial p_i}{\partial b_j}\right] - \frac{b_j}{\tau}$$

$$= \underbrace{\sum_{i:\, j=h_i}(Y_i - p_i)}_{\text{home games}} - \underbrace{\sum_{i:\, j=a_i}(Y_i - p_i)}_{\text{away games}} - \frac{b_j}{\tau}$$